TWO-WEEK COLLABORATIVE DESIGN CHALLENGE

BioSense PH

An integrated biomedical signal-monitoring and decision-support demonstrator for community health settings in the Philippines

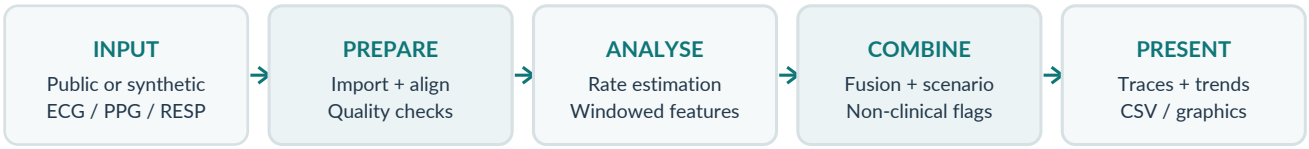
ISSUED 21 October 2026 | SUBMISSION 4 November 2026 | FORMAT One class-wide system | Eight teams

01 / The challenge

Design and demonstrate a **working, computer-based biomedical monitoring prototype** that imports physiological recordings, estimates selected vital-signal features, checks data quality, displays interpretable trends, and generates transparent **simulation-only** alerts. The target setting is a resource-conscious community or primary-care environment; the output is an educational engineering prototype, **not a medical device**.

This is **one coordinated project**, not eight unrelated submissions. Each team will implement an assigned module, agree on common interfaces, and contribute to one final demonstration.

02 / Required system architecture



One shared demonstration: consistent data interfaces, traceable methods and visible testing

03 / Measurable outcomes

- **Programming:** import biomedical time series, manipulate arrays/data frames and produce reproducible outputs using Python.
- **Biomedical interpretation:** estimate heart and breathing rates; distinguish directly measured/reference values from inferred features.
- **Engineering integration:** combine eight modules through a shared data specification and a single launch procedure.
- **Evaluation:** show quantitative comparisons, at least three stress-test scenarios, plots with units, and documented limitations.

Scope boundary. No sensors, purchases, internet-connected services, live patient monitoring, or clinical diagnosis are required. Publicly licensed recordings and synthetic alterations are sufficient. **Never use identifiable personal health records.**

WORK PACKAGE DISTRIBUTION

Eight connected technical teams

Each team must deliver executable code, a defined input/output interface, one explanatory visual and evidence of at least two checks. Coordinate early: modules will be assessed both individually and as a shared system.

TEAM 01 Data import & alignment

Read two BIDMC recordings or approved synthetic equivalents. Verify sampling rate, timestamps and signal availability; provide a common loader.

Evidence: Load both records and display a compact data summary.

TEAM 02 ECG processing

Visualize, optionally filter, and detect cardiac peaks to estimate windowed heart rate (HR). Compare estimates with reference HR.

Evidence: ECG plot with detected peaks and an error summary.

TEAM 03 PPG processing

Estimate pulse rate (PR) from photoplethysmography and flag unstable/poor-quality windows. Compare PR with reference PR.

Evidence: PPG plot, pulse markers and a validation result.

TEAM 04 Respiratory analysis

Analyse respiration waveform to estimate breaths per minute (RR). Compare predicted RR with a reference series or annotations.

Evidence: Respiratory trace, detected breaths and error.

TEAM 05 Signal-quality assurance

Test handling of noise, flatlines, missing samples, abnormal spikes and unusable windows; emit explicit quality labels.

Evidence: Three modified examples with pass/fail evidence.

TEAM 06 Feature fusion & flags

Combine windowed HR, PR, RR and reference SpO2 into a timeline. Implement editable demo thresholds and no-data rules.

Evidence: Show clean, caution and missing-data scenarios.

TEAM 07 Dashboard & reporting

Build one usable launch screen (desktop/notebook/Streamlit) with signals, trends, units, legend, warnings and exports.

Evidence: A single demonstration page and exportable results.

TEAM 08 Integration & verification

Connect all modules, test the end-to-end launch, prepare reproducible setup steps, and coordinate system-level verification.

Evidence: One command launches a tested end-to-end demo.

Shared responsibility: Every team shall explain its biomedical relevance, provide a minimum of two test cases, retain source code, and participate in the integrated demonstration. Team 01 publishes the common data format by Day 2; Team 08 checks compatibility continuously.

TECHNICAL SPECIFICATION

Build requirements and verification

04 / Tools, datasets and formats

Required: Python 3 with NumPy, Pandas and Matplotlib; SciPy is recommended. Jupyter Notebook or a simple Python application is acceptable. Streamlit is optional. The system must run without a paid service.

Reference data: Prefer PhysioNet BIDMC PPG and Respiration Dataset (v1.0.0). Its CSV signals include ECG, PPG and impedance respiration at 125 Hz; companion numeric records provide physiological reference values. Use **at least two distinct recordings** (for example, 01 and 02) or well-documented synthetic equivalents. [1]

Data interface: Team 01 defines a common internal waveform object with record identifier, time in seconds, sampling frequency and signal arrays. Every analytical team exports a simple feature table in the agreed schema below.

FIELD	MEANING / ALLOWED VALUES
record_id	Dataset or scenario identifier (e.g., bidmc-01)
start_s / end_s	Start/end of a non-overlapping 30-second window
feature / value	HR, PR, RR, SpO2 reference, or scenario flag; numeric estimate where relevant
unit / quality	bpm, breaths/min, %, or label; quality = OK, LOW, MISSING

05 / Minimum analysis and validation

- **Signal logic:** disclose filtering/peak-detection choices. If N valid successive peaks occur at times t1 and tN (seconds), an illustrative rate estimate is 60(N - 1)/(tN - t1). Apply this for suitable cardiac or respiratory peaks; check edge cases.
- **Fair comparison:** align an estimate with reference values from the same 30-second interval. Report mean absolute error (MAE), number of evaluated windows, and valid-window coverage; show at least one error plot.
- **Scenario testing:** use (A) unmodified data, (B) deliberately injected noise or spikes, and (C) removed or flatline segments. Explain how quality flags affect plots and alert rules.
- **Transparent flags:** instructor-configurable demonstration thresholds only. Never call an educational alert a diagnosis or an emergency recommendation. SpO2 is a reference input, not something derived from single-channel PPG.

06 / Integrity, safety and reproducibility

- Use de-identified, appropriately licensed public data or synthetic data only; credit the dataset and method sources. No named patient records, identifiers or recordings from clinical placements.
- Make data lineage clear: raw signal vs processed signal vs calculated feature vs provided reference vs simulated flag.
- Keep reproducible execution instructions, a dependency list and a consistent set of example inputs; include known failures and realistic limitations.

Data-protection requirement. Health information is sensitive personal information under the Philippine Data Privacy Act of 2012 (Republic Act No. 10173). This assignment explicitly avoids processing identifiable health information. [3]

DELIVERY MANAGEMENT

Fourteen-day execution plan

The brief is issued on **Wednesday, 21 October 2026**; the fourteen-day implementation window runs from **22 October to 4 November 2026**. Each team must make its progress visible during integration checks; the entire platform must be operational on the due date.

PHASE	DATES	EXPECTED ACTION	CHECKPOINT
DAYS 01-02	22-23 Oct	Form the eight teams; appoint leads; choose two records; approve data schema, scope and interfaces.	Architecture sketch and schema agreed.
DAYS 03-05	24-26 Oct	Build import pipeline, initial signal functions and mock outputs; meet for integration checkpoint on Day 5.	First runnable module from each team.
DAYS 06-08	27-29 Oct	Complete primary methods, plotting, data quality flags, fusion logic and dashboard wiring.	Feature tables + measurable baseline results.
DAYS 09-11	30 Oct-1 Nov	Integrate all eight components; run normal, noisy and missing-data demonstrations; fix interface errors.	Full prototype with verification notes.
DAYS 12-13	2-3 Nov	Finish graphs, report, slides, references and dry-run; verify clean-install instructions.	Submission candidate with no missing files.
DAY 14	4 Nov	Conduct final quality check and submit by Wednesday, 4 November 2026, 11:59 p.m. Philippine Standard Time.	One class-wide submission; presentation ready.

07 / Final submission package

Submit **one compressed project folder** for the entire class, containing the following six items. Use clear filenames without spaces or special characters where needed for execution; the final class archive may use a readable project title.

- **Executable prototype:** all eight Python modules, the single launch script and an environment/dependency list.
- **Reproducibility guide:** a concise README specifying installation, dataset URLs, required commands, source attribution and expected output.
- **Engineering report:** one consolidated PDF, approximately 10-15 pages (excluding references/appendices): motivation, architecture, modules, methods, validation, limitations and contributions.
- **Presentation:** one unified slide deck of 12-15 slides, including the architecture, three central results, integration outcome and limitations.
- **Evidence:** plots with axes and units, test results, performance/error table, and screenshots or a short local demonstration recording.
- **Team contribution matrix:** name, team, assigned function, code/report contribution and the evidence used to verify it.

ASSESSMENT AND PRESENTATION

Demonstrate, defend, and reflect

08 / Presentation protocol

The class will deliver **one integrated presentation**: **18 minutes** for the technical presentation and live demonstration, followed by **7 minutes** for questions. All eight teams must be represented in the technical explanation; the order and speakers should be planned to avoid repetition. The exact presentation session will be announced separately.

- **Opening (2 min):** biomedical need, problem definition and architecture.
- **Methods and programming (8 min):** team modules, shared interfaces and one representative piece of code or pseudocode.
- **Evidence (5 min):** signal plots, reference comparisons, noisy/missing-data tests and the functioning dashboard.
- **Limitations and closure (3 min):** accuracy, reproducibility, privacy, risks and realistic future improvement.

09 / Evaluation rubric

ASSESSMENT DIMENSION	MARKS
System design, architecture and module integration	15
Technical implementation appropriate to the assigned package	20
Quantitative validation, robustness and data quality	20
Code clarity, documentation and reproducible execution	15
Data visualisation, interface and live demonstration	10
Consolidated report, references and ethical interpretation	10
Presentation, technical defence and contribution evidence	10
TOTAL	**100**

Acceptance gate: The README must enable a reviewer to execute the platform, inspect three physiological plots, reproduce three test scenarios, and verify output metrics. A slide-only concept, unexplained external calls, or unsupported clinical claims do not meet the requirement.

10 / Approved references and starting points

[1] Pimentel et al., *BIDMC PPG and Respiration Dataset*, PhysioNet, v1.0.0, 2018. https://physionet.org/content/bidmc/1.0.0/ (DOI: 10.13026/C2208R).
[2] M. A. F. Pimentel et al., "Towards a Robust Estimation of Respiratory Rate from Pulse Oximeters," *IEEE Transactions on Biomedical Engineering*, 2016. DOI: 10.1109/TBME.2016.2613124.
[3] Republic of the Philippines, Republic Act No. 10173, *Data Privacy Act of 2012*. https://privacy.gov.ph/data-privacy-act-/.

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